

Clinical Trial Protocol

Protocol ID: PROT-2026-001

Title: A Phase II, Randomized, Double-Blind, Placebo-Controlled Study Evaluating the Efficacy and Safety of Investigational Agent DX-410 in Adult Patients with Type 2 Diabetes Mellitus

1. Study Overview

This study evaluates the glycemic efficacy and safety of DX-410 versus placebo, administered once daily over a 24-week treatment period, in adult patients with inadequately controlled Type 2 Diabetes Mellitus (T2DM) despite stable background therapy.

2. Inclusion Criteria

- 2.1 Male or female, age 18 to 75 years, inclusive, at time of screening.
- 2.2 Confirmed diagnosis of Type 2 Diabetes Mellitus for at least 6 months prior to screening.
- 2.3 Estimated glomerular filtration rate (eGFR) > 60 mL/min/1.73m² at screening, calculated using the CKD-EPI equation.
- 2.4 Glycated hemoglobin (HbA1c) between 7.0% and 10.0%, inclusive, at screening.
- 2.5 Body Mass Index (BMI) between 25 and 40 kg/m², inclusive.
- 2.6 Stable dose of metformin (≥ 1500 mg/day or maximum tolerated dose) for at least 8 weeks prior to screening.
- 2.7 Willing and able to provide written informed consent and comply with study visit schedule.

3. Exclusion Criteria

- 3.1 HbA1c $> 10.0\%$ at screening.
- 3.2 History of Type 1 Diabetes Mellitus or history of diabetic ketoacidosis.
- 3.3 History of acute or chronic pancreatitis.
- 3.4 Currently pregnant, breastfeeding, or planning pregnancy during the study period.
- 3.5 Use of a GLP-1 receptor agonist within 14 days prior to screening (washout period required: 14 days).
- 3.6 eGFR ≤ 60 mL/min/1.73m² at screening.
- 3.7 Uncontrolled hypertension, defined as systolic blood pressure > 160 mmHg or diastolic > 100 mmHg at screening.
- 3.8 Known hypersensitivity to DX-410 or its excipients.

4. Study Design

This is a multi-center, randomized, double-blind, placebo-controlled, parallel-group study. Eligible participants will be randomized 1:1 to receive either DX-410 or matching placebo, administered once daily for 24 weeks, in addition to background metformin therapy.

5. Screening & Eligibility Assessment Schedule

Assessment	Screening (Day -14 to -1)	Baseline (Day 0)
Informed Consent	X	
Medical History	X	
HbA1c	X	X
eGFR / Renal Panel	X	
Pregnancy Test (if applicable)	X	X
Vital Signs	X	X
Concomitant Medication Review	X	X

6. Primary Endpoint

Change from baseline in HbA1c at Week 24, compared between the DX-410 and placebo arms.

7. Safety Monitoring

Adverse events will be monitored throughout the study. Participants meeting any exclusion criterion identified after enrollment will be flagged for immediate investigator review and possible discontinuation, consistent with 21 CFR Part 11 documentation requirements for all eligibility determinations.